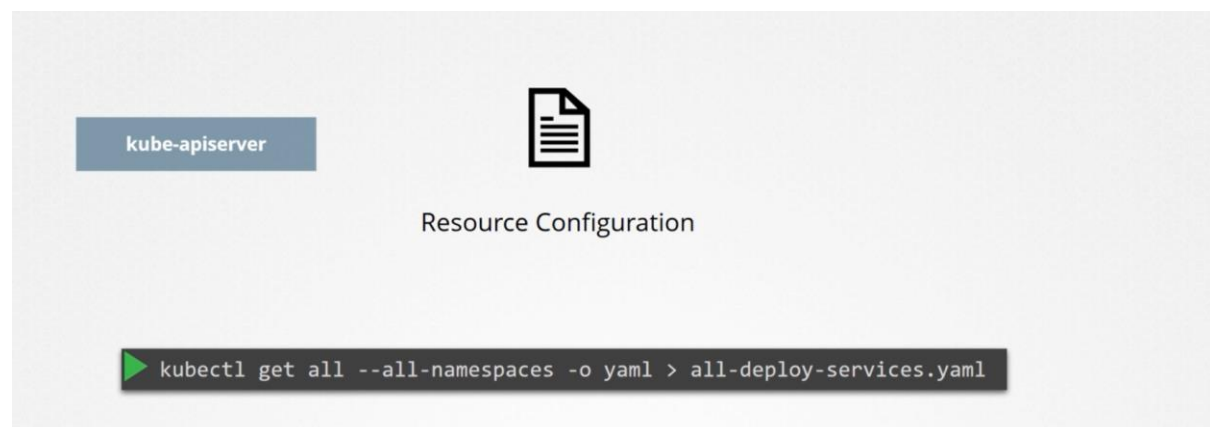
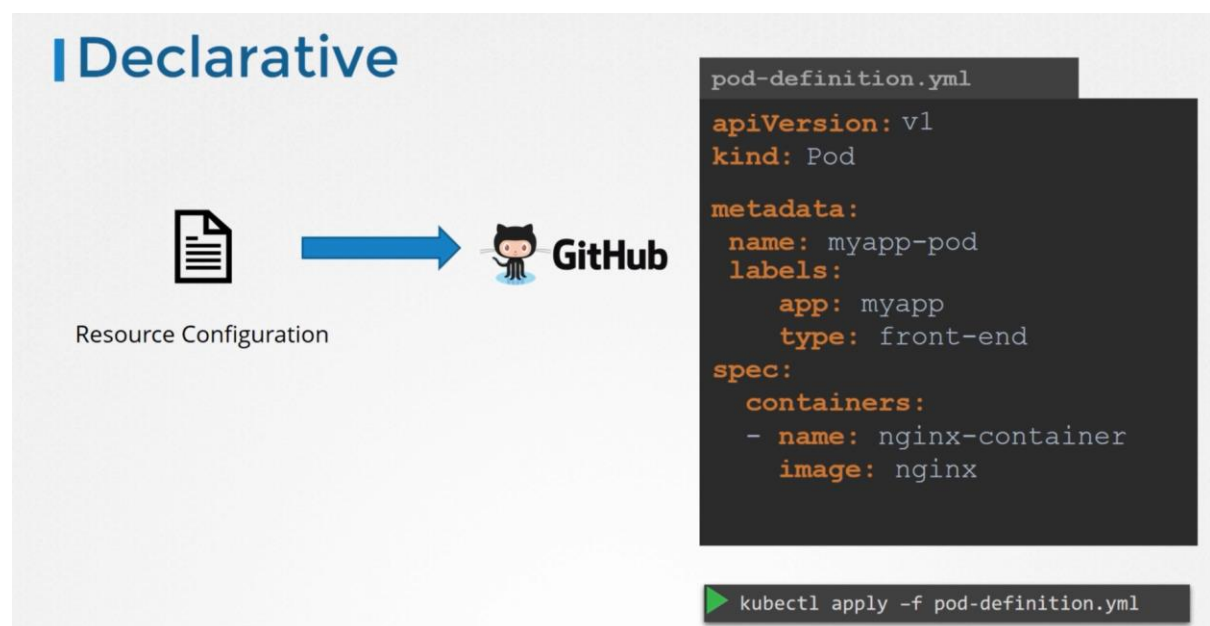


## Backup and Restore:

### Backup Candidates



## Declarative



## ETCD cluster

ETCD store information about state of the cluster, cluster info stored in etcd

**etcd**- entire cluster state (metadata) available in etcd database



ETCD cluster is hosted on master node, while configuring etcd specify the location where all the data will be stored.

**ETCD also comes with built in snapshot solution.** You can take snapshot of etcd database.

**To** restore the cluster from this backup on later time, first stop the kube-apiserver service.

ETCD is deployed as static pod in K8s cluster in kube-system ns.

### Who communicate with etcd database?

api-server, how to communicates with etcd database directly with the help of client utility **etcdctl**.

Only api server, can access etcd component by default, if we have to access separately then you have to install separate utility client on your box is called **etcdctl**. etcdctl is a client to access etcd server.

### step1: Install etcdctl

```
export RELEASE="3.3.13"
```

```
wget https://github.com/etcd-io/etcd/releases/download/v${RELEASE}/etcd-v${RELEASE}-linux-amd64.tar.gz
```

```
tar xvf etcd-v${RELEASE}-linux-amd64.tar.gz
```

```
cd etcd-v${RELEASE}-linux-amd64
```

```
sudo mv etcdctl /usr/local/bin
```

```
etcdctl --version
```

To communicate with etcd, client must know where etcd db is, must know the ip and port of etcd db.

```
kubectl describe pod etcd-master -n kube-system
```

```
Command:
etcd
--advertise-client-urls=https://172.17.0.90:2379
--cert-file=/etc/kubernetes/pki/etcd/server.crt
--client-cert-auth=true
--data-dir=/var/lib/etcd
--initial-advertise-peer-urls=https://172.17.0.90:2380
--initial-cluster=master=https://172.17.0.90:2380
--key-file=/etc/kubernetes/pki/etcd/server.key
--listen-client-urls=https://127.0.0.1:2379,https://172.17.0.90:2379
--listen-metrics-urls=http://127.0.0.1:2381
--listen-peer-urls=https://172.17.0.90:2380
--name=master
--peer-cert-file=/etc/kubernetes/pki/etcd/peer.crt
--peer-client-cert-auth=true
--peer-key-file=/etc/kubernetes/pki/etcd/peer.key
--peer-trusted-ca-file=/etc/kubernetes/pki/etcd/ca.crt
--snapshot-count=10000
--trusted-ca-file=/etc/kubernetes/pki/etcd/ca.crt
```

Since etcd is secure server, it does not allow any http communication must go through the https, it has its own certificates

/etc/kubernetes/pki/etcd

ca.crt, server.crt, server.key

end point is server ip address

```
ETCDCTL_API=3 etcdctl snapshot save -h
```

```
ETCDCTL_API=3 etcdctl --endpoints="" --cacert="" --cert="" --key="" snapshot
save /tmp/etcd-backup.db
```

```
ETCDCTL_API=3 etcdctl --endpoints="" --cacert="" --cert="" --key="" -w=table
snapshot status /tmp/etcd-backup.db
```

## Validate:

```
ETCDCTL_API=3 etcdctl snapshot member list --  
cacert=/etc/kubernetes/pki/etcd/ca.cert --  
cert=/etc/kubernetes/pki/etcd/server.crt --  
key=/etc/kubernetes/pki/etcd/server.key --endpoints=127.0.0.1:2379
```

## How to restore your backup?

Steps of restore of ETCD db?

```
ETCDCTL_API=3 etcdctl snapshot restore -h
```

```
ETCDCTL_API=3 etcdctl --data-dir= /var/lib/etcd-from-backup snapshot restore  
/tmp/etcd-backup.db
```

```
cat /etc/kubernetes/manifests/etcd.yaml # update the hostPath /var/lib/etcd-  
>/var/lib/etcd-from-backup
```